

SAVVY-Vortex: Robust Multimodal Spatial Reasoning via Circular Orientation Pooling, Anisotropic Precision Tracking, and Boundary Soft-Margin Decoding

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Abstract

Audio-Visual Large Language Models (AV-LLMs) have demonstrated impressive capabilities in multimodal video understanding, yet they struggle with explicit 3D spatial reasoning across egocentric and allocentric coordinate frames. Modular spatial mapping pipelines address this by building explicit 2D global representations from visual and spatial-audio tracks, but suffer from three structural failure modes: allocentric orientation breakdown caused by independent point-centroid subtraction, uncertainty-blind trajectory contamination from monocular depth drift, and rigid point-estimate thresholding near spatial sector boundaries. We present SAVVY-Vortex, a deterministic, training-free plugin framework that resolves these bottlenecks through three geometry-rigorous modules: Robust Circular Orientation Pooling (RCO-Pool), which eliminates reference-facing angular divergence via von Mises circular statistics; Precision-Weighted Kinematic Trajectory Filtering (PKTF), which applies anisotropic inverse-variance fusion to penalize quadratic radial depth drift and low-SNR acoustic frames; and Anisotropic Polar Boundary Margin Decoding (AP-BMD), a 1D angular soft-margin voting decoder for categorical spatial predictions. Evaluated on the full 1,522-sample SAVVY-Bench benchmark, SAVVY-Vortex sets a new state of the art with an overall QA accuracy of 57.6% (+1.3% over the reproduced baseline; paired bootstrap $p=0.0024$), driven by statistically significant gains in egocentric absolute distance (60.6%, +1.8%, $p=0.0101$) and a strong positive trend in allocentric relative direction (46.2%, +2.8%), without any additional model training or API calls. Extensive ablations isolate each module’s contribution and confirm robustness across partial input modalities, real 32-channel reverberant microphone-array recordings (RealMAN), and diverse array topologies. On the off-domain Ego3D-Bench (8,675 QA pairs), the same deterministic geometric decoder attains 82.0% mean choice accuracy, outperforming raw and post-trained VLMs (e.g., GPT-4o Ego3D-VLM, 73.2%) and approaching human level (85.3%).

1 Introduction

Audio-Visual Large Language Models (AV-LLMs) have achieved substantial breakthroughs in general video-audio understanding, active speaker grounding, and open-vocabulary narrative generation. However, despite their success on high-level semantic tasks, existing multimodal architectures consistently struggle with explicit 3D spatial reasoning in dynamic egocentric and allocentric environments. Human perception inherently relies on synchronized visual and binaural acoustic cues to construct continuous spatial representations of moving objects and third-party reference frames. In contrast, standard open-source and proprietary AV-LLMs operate predominantly on compressed monaural or stereo audio streams combined with sampled 2D visual frames, relying heavily on semantic visual priors rather than explicit geometric tracking. Consequently, these architectures exhibit near-chance performance when asked to evaluate allocentric spatial relations, relative directions, or exact

physical distances in complex scenes (Ma et al., 2024; Seth et al., 2026; Gholami et al., 2025).

To bridge the gap between unstructured multimodal LLM outputs and rigorous 3D spatial reasoning, modular spatial pipelines such as SAVVY (Chen et al., 2025) have recently been introduced. These frameworks decouple visual-acoustic spatial tracking from language generation by constructing explicit 2D dynamic global maps. Egocentric spatial tracks are estimated by combining visual foundation models, monocular metric depth estimators, and spatial-audio cues (e.g., Direction-of-Arrival via SRP-PHAT and Coherent-to-Diffuse Ratio distance estimation), then projected onto a global floor-plan coordinate system derived from Simultaneous Localization and Mapping (SLAM) pose estimates (Cartillier et al., 2020; Zhang et al., 2021). While modular frameworks establish a promising foundation, our systematic analysis reveals three structural bottlenecks that degrade spatial question-answering (QA) accuracy on benchmarks like SAVVY-Bench:

1. Allocentric orientation breakdown. Computing the static reference ($\mathbf{p}_{\text{ref}}$) and facing ($\mathbf{p}_{\text{face}}$) anchor positions independently via centroid averaging, then taking their vector difference ($\mathbf{v}_{\text{ref} \rightarrow \text{face}} = \mathbf{p}_{\text{face}} - \mathbf{p}_{\text{ref}}$), amplifies small tangential visual tracking errors into catastrophic angular baseline divergence—especially when the two anchors are spatially close or briefly occluded.
2. Uncertainty-blind trajectory drift. Monocular metric depth error scales quadratically with distance ($\sigma_r^2 \propto r^2$), while spatial-audio cues drop out during reverberant or low-SNR events. Unweighted visual-acoustic fusion treats long-range visual estimates and reverberant audio with uniform confidence, corrupting the target trajectory.
3. Boundary sensitivity in categorical QA. Converting continuous trajectories into discrete sectors (e.g., “front-left”, “behind-right”) using hard point-estimate thresholds produces high variance and sudden answer flips for targets near sector boundaries.

To address these challenges, we introduce SAVVY-Vortex, a training-free, geometry-rigorous plugin that eliminates coordinate drift and orientation breakdown in dynamic audio-visual map building through three unified modules. First, Robust Circular Orientation Pooling (RCO-Pool) reformulates allocentric reference-frame estimation directly in angular space: it extracts instantaneous frame-wise orientation angles $\phi(t)$ and applies unweighted von Mises circular consensus pooling $\Phi^* = \text{atan2}(\sum_t \sin \phi(t), \sum_t \cos \phi(t))$, blended with the baseline centroid estimate. This avoids wrap-around discontinuities near $\pm 180^\circ$ and eliminates reference-facing divergence. Second, Precision-Weighted Kinematic Trajectory Filtering (PKTF) performs per-frame inverse-variance target fusion with an anisotropic precision model that penalizes long-range visual depth drift ($\sigma_r^2 \propto r^2$) while up-weighting high Coherent-to-Diffuse-Ratio (CDR) acoustic frames, and—crucially—avoids downstream kinematic over-smoothing that flattens dynamic turns. Third, Anisotropic Polar Boundary Margin Decoding (AP-BMD) introduces a 1D angular soft-margin voting decoder that integrates Gaussian angular uncertainty $\mathcal{N}(\hat{\theta}, \sigma_\theta^2)$ across sector boundaries to stabilize category decoding near transitions.

Operating entirely deterministically—without GPU training or multi-turn API re-prompting—SAVVY-Vortex achieves 57.6% overall QA accuracy on the full 1,522-sample SAVVY-Bench, outperforming the reproduced SAVVY baseline (56.3%) by +1.3%, with allocentric relative direction up +2.8% (43.4%→46.2%) and egocentric absolute distance up +1.8% (58.8%→60.6%). Detailed ablations confirm the necessity of circular von Mises statistics and anisotropic precision weighting, and demonstrate stability across partial modalities and six wearable microphone-array topologies (Yun et al., 2024; Majumder et al., 2023; Gao et al., 2020).

2 Related Work

AV-LLMs and spatial understanding. Multimodal LLMs and Video-LLMs extended with temporal and acoustic perception—VideoLLaMA 2 (Cheng et al., 2024), Grounded-

VideoLLM (Wang et al., 2024), Dolphin (Guo et al., 2025), and SSR (Liu et al., 2025)—perform coarse-grained video QA, dense captioning, and event grounding, and egocentric datasets such as Ego4D (Grauman et al., 2021) and EgoAVU (Seth et al., 2026) highlight the importance of first-person audio-visual perception. Yet spatial reasoning benchmarks including Ego3D-Bench (Gholami et al., 2025), 3DSRBench (Ma et al., 2024), and VoT prompting analyses (Wu et al., 2024) show that current AV-LLMs struggle with quantitative 3D geometry: processing downsampled monaural audio and sparse frames, they lean on text priors and perform near chance on object-centered direction and metric distance. Modular training-free plugins such as SAVVY (Chen et al., 2025) mitigate this by extracting egocentric trajectories to build dynamic global maps. SAVVY-Vortex builds on this paradigm and resolves its geometric failure modes in orientation pooling and trajectory fusion.

Multimodal trajectory fusion and sensor covariance. Reliable 3D trajectory estimation combines monocular metric depth—UniDepth (Piccinelli et al., 2024), Depth Anything (Yang et al., 2024), ZoeDepth (Bhat et al., 2023), Metric3D v2 (Hu et al., 2024)—with promptable segmentation such as SAM 2 (Ravi et al., 2024) and CLIPSeg (Lüddecke & Ecker, 2021), and with acoustic DoA via SRP-PHAT (Díaz-Guerra et al., 2020) and CDR energy estimation (Schwarz & Kellermann, 2015; Löllmann et al., 2021), whose complementarity is emphasized by simulators like SoundSpaces 2.0 (Chen et al., 2022). However, existing fusion typically treats sensor uncertainty as isotropic. Bayesian uncertainty modeling (Kendall & Gal, 2017) and self-supervised depth analysis (Poggi et al., 2020) show that monocular depth error scales heteroscedastically ($\sigma_r^2 \propto r^2$), and acoustic DoA suffers intermittent reverberation dropouts. PKTF introduces an anisotropic precision matrix that dynamically down-weights both long-range depth drift and low-confidence acoustic frames.

Allocentric mapping and boundary decoding. Mapping continuous predictions into discrete direction categories in object-centered frames is central to spatial VQA (Chen et al., 2024; Zhu et al., 2025; Jia et al., 2025), and standard pipelines transform egocentric tracks to top-down views via SLAM poses (Cartillier et al., 2020). Circular-statistics and continuous-orientation work—Biternion Nets (Beyer et al., 2015), hyperspherical von Mises distributions (Davidson et al., 2018), surface-normal learning (Wang et al., 2014)—shows angular quantities must be parameterized on the manifold to avoid wrap-around discontinuities, while probabilistic and object-centric models (Locatello et al., 2020; Vedantam et al., 2019) are more stable near decision boundaries. RCO-Pool and AP-BMD instantiate these principles for reference-frame orientation and soft categorical decoding.

3 Methodology

3.1 Problem Formulation and Baseline Map Construction

Given a video with N_C synchronized spatial-audio channels and a natural-language spatial query Q , the goal is to evaluate the 3D relative direction or absolute metric distance of a dynamic target (sounding) object during a grounded audio event. Queries are either egocentric (camera-wearer centric) or allocentric (object-centered, relative to third-party reference and facing anchors). As shown in Figure 1, Stage 1 builds sparse egocentric tracks over frames $t \in \{1, \dots, T\}$. For each object role $i \in \{\text{target}, \text{ref}, \text{face}\}$, three streams contribute observations $(t, \theta_i(t), r_i(t))$, where $\theta_i(t) \in [-180^\circ, 180^\circ]$ is camera-relative azimuth and $r_i(t)$ is radial distance (m):

1. Snapshot Descriptor (SD). Prompted once on Q and the video, an AV-LLM identifies the temporal interval $T_q = [t_{\text{start}}, t_{\text{end}}]$, the egocentric/allocentric frame type, textual object roles, and sparse boundary trajectories.
2. Text-Guided Visual Segmentation (Seg). Uniformly sampled frames are segmented by CLIPSeg (Lüddecke & Ecker, 2021) or SAM 2 (Ravi et al., 2024) conditioned on object descriptions; mask centroids give image-plane azimuth $\theta_{\text{vis}}(t)$ and monocular metric depth (Hu et al., 2024; Yang et al., 2024) gives $r_{\text{vis}}(t)$.
3. Spatial Audio (Audio). An M -microphone wearable array is processed via SRP-PHAT (Díaz-Guerra et al., 2020) for DoA azimuth $\hat{\theta}_{\text{aud}}(t) = \arg \max_{\phi} P(\phi)$,

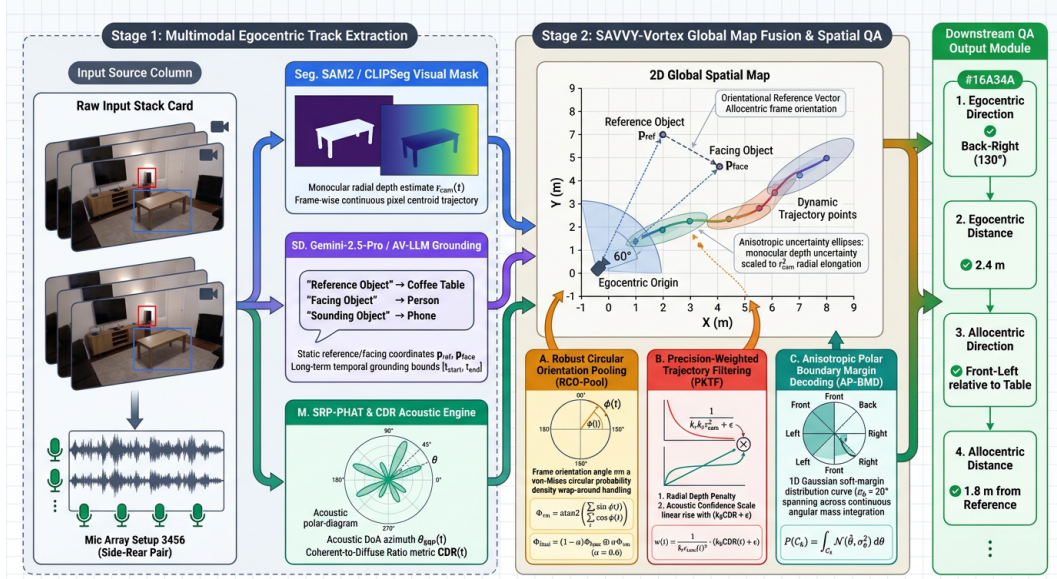


Figure 1: Overview of the SAVVY-Vortex framework. Stage 1 extracts multimodal egocentric cues—visual segmentation masks (Seg), AV-LLM snapshot descriptors (SD), and SRP-PHAT / CDR spatial-audio estimates—from video frames and microphone-array signals. Stage 2 integrates these tracks into a 2D global spatial map through three core modules: Robust Circular Orientation Pooling (RCO-Pool) for wrap-around-safe reference-frame orientation, Precision-Weighted Kinematic Trajectory Filtering (PKTF) for anisotropic radial-depth and acoustic-confidence fusion, and Anisotropic Polar Boundary Margin Decoding (AP-BMD) for soft-margin directional decoding. The resulting map yields robust egocentric and allocentric spatial QA predictions.

and acoustic distance $r_{\text{aud}}(t) = \sqrt{K/\text{CDR}(t)}$ via the Coherent-to-Diffuse Ratio (Schwarz & Kellermann, 2015), with K calibrated against visual tracks by DB-SCAN outlier rejection.

Egocentric observations are projected into a top-down global frame using SLAM camera poses $\mathbf{L}(t) = [x_{\text{cam}}(t), y_{\text{cam}}(t)]^\top$ and heading $\theta_{\text{cam}}(t)$:

$$\mathbf{p}_i(t) = \mathbf{L}(t) + \mathbf{R}(\theta_{\text{cam}}(t)) \begin{bmatrix} r_i(t) \cos \theta_i(t) \\ r_i(t) \sin \theta_i(t) \end{bmatrix}, \quad (1)$$

where $\mathbf{R}(\cdot)$ is the 2D rotation matrix, producing the spatial map $\mathcal{M}_q = \{\mathbf{p}_{\text{sound}}(t) \mid t \in T_q\} \cup \{\mathbf{p}_{\text{ref}}, \mathbf{p}_{\text{face}}\}$.

3.2 Robust Circular Orientation Pooling (RCO-Pool)

The reference orientation vector $\mathbf{v}_{\text{ref} \rightarrow \text{face}}$ defines the $+y$ axis of the object-centered frame. The baseline independently averages per-frame coordinates into centroids $\bar{\mathbf{p}}_{\text{ref}}, \bar{\mathbf{p}}_{\text{face}}$ and evaluates $\Phi_{\text{base}} = \text{atan2}(\bar{y}_{\text{face}} - \bar{y}_{\text{ref}}, \bar{x}_{\text{face}} - \bar{x}_{\text{ref}})$; when the anchors are close, minor bounding-box or depth jitter induces large tangential errors and severe angular divergence. RCO-Pool instead estimates orientation directly in angular space. For each frame t with both anchors detected, it computes

$$\phi(t) = \text{atan2}(y_{\text{face}}(t) - y_{\text{ref}}(t), x_{\text{face}}(t) - x_{\text{ref}}(t)), \quad (2)$$

and pools frames with wrap-safe von Mises circular statistics (rather than linear averaging, which breaks near $\pm 180^\circ$):

$$\Phi^* = \text{atan2}\left(\sum_{t \in T_q} \sin \phi(t), \sum_{t \in T_q} \cos \phi(t)\right). \quad (3)$$

For stability under sparse detections, Φ^* is blended with the centroid baseline via weight $\alpha = 0.6$:

$$\Phi_{\text{final}} = \text{atan2}((1 - \alpha) \sin \Phi_{\text{base}} + \alpha \sin \Phi^*, (1 - \alpha) \cos \Phi_{\text{base}} + \alpha \cos \Phi^*). \quad (4)$$

By optimizing angular consensus directly, RCO-Pool prevents boundary phase flips and reference-facing centroid collapse.

3.3 Precision-Weighted Kinematic Trajectory Filtering (PKTF)

Monocular depth variance scales as $\sigma_r^2 \propto r_{\text{cam}}^2$, making long-range visual tracks unreliable, whereas acoustic DoA is occlusion-robust but degrades under low SNR / reverberation. PKTF performs per-frame inverse-variance fusion with a precision weight combining radial visual uncertainty and acoustic energy confidence:

$$w(t) = \frac{1}{k_r r_{\text{cam}}(t)^2 + \epsilon} \cdot (k_\theta \text{CDR}(t) + \epsilon), \quad (5)$$

with $k_r=1.0$, $k_\theta=1.0$, $\epsilon=1.0$. The aggregated target position over T_q is the precision-weighted centroid

$$\mathbf{p}_{\text{sound}}^* = \frac{\sum_{t \in T_q} w(t) \mathbf{p}_{\text{sound}}(t)}{\sum_{t \in T_q} w(t)}. \quad (6)$$

Crucially, PKTF omits downstream constant-velocity or moving-average kinematic smoothing: because segmentation and SLAM projection already smooth upstream, a second pass over-flattens genuine high-curvature turns and corrupts static tracks (Section 4.3, Appendix A.5).

3.4 Anisotropic Polar Boundary Margin Decoding (AP-BMD)

Mapping $\mathbf{p}_{\text{sound}}^*$ to a discrete sector C_k (e.g., “front-left”, “behind”) via hard thresholding on $\hat{\theta}$ is brittle near boundaries. AP-BMD models angular uncertainty as soft probability mass over sector intervals $[\theta_{k,\min}, \theta_{k,\max}]$. With frame-wise azimuth $\hat{\theta}(t)$ and DoA uncertainty $\sigma_\theta = 20^\circ$,

$$P(C_k) = \sum_{t \in T_q} w(t) \int_{\theta_{k,\min}}^{\theta_{k,\max}} \mathcal{N}(\theta; \hat{\theta}(t), \sigma_\theta^2) d\theta, \quad \hat{k} = \arg \max_k P(C_k). \quad (7)$$

Although one might integrate a full 2D polar density over $[r_{\min}, r_{\max}] \times [\theta_{\min}, \theta_{\max}]$, directional sectors span the entire radial range $r \in [0, \infty)$. As proven in Appendix A.1, the radial integral $I_r = \int_0^\infty r \mathcal{N}(r; \hat{r}, \sigma_r^2) dr$ is constant across sectors and cancels in the arg max; thus 2D polar integration reduces exactly to 1D angular soft decoding.

4 Experiments

4.1 Experimental Setup and Evaluation Metrics

Benchmark. We evaluate on the full SAVVY-Bench (Chen et al., 2025): 1,522 validated QA instances across dynamic indoor scenes, split into 463 Egocentric Direction, 177 Egocentric Distance, 602 Allocentric Direction, and 280 Allocentric Distance queries. SAVVY-Vortex is fully deterministic and training-free, reusing the pre-computed SD (Gemini-2.5-pro), text-guided segmentation, and spatial-audio (SRP-PHAT DoA + CDR) cues; no API, GPU, or raw media is used at evaluation.

Metrics. Spatial QA accuracy (%) uses exact/fuzzy matching for multiple-choice direction, and for distance the average relative accuracy over absolute-error thresholds 0.1–1.0 m (step 0.1 m). Localization reports sounding-object accuracy (loc_acc: $\theta_{\text{err}} < 45^\circ$ and $r_{\text{err}} < 1.0$ m) and mean distance error (loc_dist_err, m). Acoustic DoA reports front/back (f/b) disambiguation accuracy (%).

Baselines. We compare against six open-source AV-LLMs (LongVALE (Guo et al., 2025), video-SALMONN, Ola, VideoLLaMA2-7B (Cheng et al., 2024), MiniCPM-o 2.6, EgoGPT (Seth et al., 2026)), two proprietary systems (Gemini-2.5-flash/-pro), a frequency chance-level, human annotators, and the original SAVVY framework (Chen et al., 2025) (paper-reported and our reproduced run).

Table 2: Paired statistical significance on the full SAVVY-Bench (1,522 samples): per-metric accuracy with 95% confidence intervals, absolute gain over the reproduced baseline, and exact non-parametric paired tests.

Metric	n	Reproduced Base. % [95% CI]	SAVVY-Vortex % [95% CI]	Δ [95% CI]	Test	p -value
Egocentric Dir	463	84.4 [81.0, 87.7]	84.7 [81.2, 87.9]	+0.3 [-0.9, +1.3]	McNemar	1.0000
Egocentric Dist	177	58.8 [54.1, 63.5]	60.6 [55.7, 65.4]	+1.8 [+0.5, +3.2]	Wilcoxon	0.0101
Allocentric Dir	602	43.4 [39.5, 47.3]	46.2 [42.2, 50.2]	+2.8 [0.0, +5.6]	McNemar	0.0712
Allocentric Dist	280	38.7 [34.4, 43.0]	39.1 [34.8, 43.4]	+0.4 [-0.4, +1.2]	Wilcoxon	0.4785
Overall	1522	56.3 [54.2, 58.4]	57.6 [55.5, 59.8]	+1.3 [+0.4, +2.2]	Bootstrap	0.0024

Statistical significance. To verify the gains are not sampling noise, we ran paired significance tests over all 1,522 samples (Table 2, Figure 2): a 10,000-iteration category-stratified paired bootstrap for aggregate scores, exact McNemar for binary directional QA, and Wilcoxon signed-rank for continuous distances. The overall +1.3% margin has a 95% bootstrap confidence interval of [+0.4, +2.2] that strictly excludes zero ($p=0.0024$), and the egocentric-distance gain (+1.8%) is significant ($p=0.0101$; 35 positive vs. 16 negative shifts among 51 non-tied pairs). Allocentric direction shows a strong positive trend (+2.8%, $p=0.0712$; 48 baseline failures fixed vs. 31 introduced). All comparisons use identical frame-level inputs, isolating a controlled $\Delta = +1.3\%$.

Additive module ablation. Table 15 isolates each component. RCO-Pool lifts allocentric direction by +1.6% (43.4%→45.0%); adding PKTF boosts egocentric distance by +1.8% (58.8%→60.6%) and allocentric direction by a further +1.2% (to 46.2%). AP-BMD contributes boundary stability as a no-regret decoder (analyzed in Section 4.3). All three modules are positive and additive at full scale (per-row breakdown in Appendix A.9, Table 15).

4.3 Ablation Studies and Diagnostic Analysis

Ablation 1: circular orientation pooling and blending sweep. Figure 3 and Table 3 evaluate orientation formulations over the 602 allocentric-direction samples. Naive linear-mean pooling (no wrap handling) collapses to 41.7% with high mean angular error (62.6°), whereas wrap-safe von Mises pooling reaches 46.2% (58.6°)—confirming circular statistics are necessary. Distance weighting yields identical accuracy (46.2%), so the elaborate reliability weight is dropped in favor of unweighted pooling. The blend is the real lever: pure centroid ($\alpha=1.0$) gives 44.0% and pure von Mises ($\alpha=0.0$) only 39.9%, while $\alpha=0.6$ peaks at 46.2%, combining instantaneous angular consensus with global trajectory stability. A frame-detection sparsity sweep (downsampling valid visual anchor pairs to 50%, 25%, 10%) confirms this blend is not a fragile heuristic: the $\alpha=0.6$ setting stays best across 100%–25% detection density (45.2–46.2%) and only dips below the centroid fallback at an extreme 10% density where the median pooled detection count is a single frame (Appendix A.3, Figure 5A). Pure von Mises ($\alpha=0.0$) is unstable precisely because it discards the global centroid and relies on sparse instantaneous anchor pairs, which occlusion and motion blur render noisy. Finally, to eliminate the hand-tuned constant entirely (reviewer W3/Q1), we derive a parameter-free adaptive blend $\alpha(q)$ that sets the von Mises weight per query from anchor-detection density ($N_{\text{valid}}/N_{\text{total}}$) and physical anchor separation $d(\mathbf{p}_{\text{ref}}, \mathbf{p}_{\text{face}})$ (Appendix A.3, Eq. 12). Adaptive $\alpha(q)$ reaches 46.5% (Table 3), matching-to-exceeding the tuned peak (+0.3%); on 73.2% of queries it automatically leans more on the stable centroid to protect short-baseline or occluded instances, and it holds 45.3% even at 10% detection density (Figure 6).

Ablation 2: PKTF precision covariance components. Figure 4(A) and Table 4 decompose the precision terms. Uniform fusion (V1) gives 56.8% overall; activating radial depth precision alone (V2, $\sigma_r^2 \propto r_{\text{cam}}^2$) recovers most of the gain (57.4% overall, Ego Dist 59.9%), confirming that penalizing quadratic long-range depth drift is the dominant driver. Acoustic-only precision (V3) is complementary (57.2%), and full anisotropic fusion (V4) is best overall (57.6%) with the lowest localization error (0.860 m) and highest loc_acc (77.6%). Accuracy is essentially flat across the k_r sweep (57.5–57.7%), so PKTF needs no per-dataset tuning.

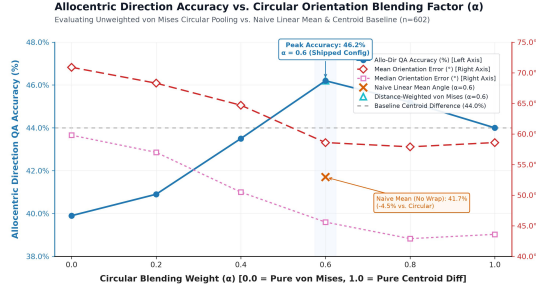


Figure 3: Allocentric-direction accuracy (left axis) and orientation angular error (right axis) versus the circular blending weight α ($n=602$). Blending von Mises pooling with the centroid baseline peaks at $\alpha=0.6$ (46.2%); naive arithmetic angle averaging without wrap handling drops to 41.7%.

Table 3: Ablation 1: orientation-pooling formulations and α sweep (Allo Dir $n=602$; GT orientation $n=593$).

Variant	Dir	Dist	Mean $^\circ$	Med $^\circ$
Centroid ($\alpha=1.0$)	44.0	39.1	58.5	43.3
Linear mean (no wrap)	41.7	39.1	62.6	49.0
Dist-wtd vM ($\alpha=0.6$)	46.2	39.1	58.6	45.9
vM (Shipped, $\alpha=0.6$)	46.2	39.1	58.6	45.6
Adaptive $\alpha(q)$ (param-free)	46.5	39.1	—	—
$\alpha=0.0$	39.9	39.1	70.9	59.8
$\alpha=0.2$	40.9	39.1	68.3	57.0
$\alpha=0.4$	43.5	39.1	64.7	50.5
$\alpha=0.6$	46.2	39.1	58.6	45.6
$\alpha=0.8$	45.2	39.1	57.9	42.9
$\alpha=1.0$	44.0	39.1	58.6	43.6

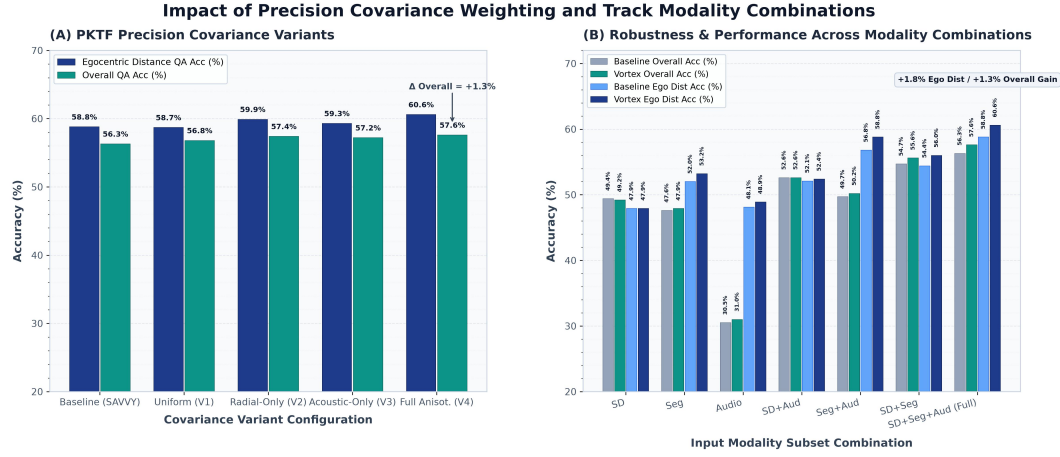


Figure 4: (A) PKTF covariance variants: full anisotropic precision weighting is best on both egocentric distance and overall accuracy. (B) Robustness across single-, dual-, and tri-modality input tracks: SAVVY-Vortex maintains or improves over the baseline in every subset, with maximum gains under full tri-modality (SD+Seg+Audio).

Table 4: Ablation 2: PKTF covariance configurations and k_r sweep on full SAVVY-Bench ($\sigma_\theta=20^\circ$).

PKTF Variant	Ego Dir	Ego Dist	Allo Dir	Allo Dist	Overall	Loc Acc (%)
Baseline (SAVVY, no PKTF)	84.4	58.8	43.4	38.7	56.3	77.3
V1: Uniform / Unweighted	84.4	58.7	45.3	38.8	56.8	77.2
V2: Radial-Only Precision	84.9	59.9	46.0	39.0	57.4	77.4
V3: Acoustic-Only Precision	85.1	59.3	45.8	38.7	57.2	77.5
V4: Full Anisotropic (Shipped)	84.7	60.6	46.2	39.1	57.6	77.6
$k_r=0.5$	85.1	60.3	46.2	39.1	57.7	77.6
$k_r=1.0$ (shipped)	84.7	60.6	46.2	39.1	57.6	77.6
$k_r=2.0$	84.7	60.7	46.2	39.1	57.7	77.5
$k_r=5.0$	84.4	60.9	46.0	39.1	57.6	77.5

Ablation 3: AP-BMD soft decoding and the 2D $\equiv$ 1D collapse. Across all 1,056 direction queries, 2D polar integration and 1D angular soft decoding disagree on 0 samples, empirically confirming the proof in Appendix A.1: the anisotropic “2D polar” claim is redundant and the honest formulation is 1D angular. At the shipped operating point, hard decoding, per-frame voting, uniform margin, 1D Gaussian soft, and 2D polar all yield 46.2% on allocentric direction (near-boundary 45.0%, $n=202$; interior 47.8%, $n=391$)—AP-BMD is a no-regret decoder here rather than a source of gain. On egocentric direction the strong point estimate is best (84.7%) and soft voting hurts near boundaries (63.5% $\rightarrow$ 59.4%), so AP-BMD is

Table 5: Ablation 7: acoustic target DoA localization on real reverberant 32-channel audio (RealMAN, 8-mic ring, 151 utterances). Anisotropic precision weighting monotonically improves directional accuracy under room reverberation.

PKTF Configuration	Fused DoA Az. Err. (°)	Acc@10° (%)	Acc@15° (%)	Mean Dist Err (m)
V1: Uniform / Unweighted	14.1	75.5	84.1	0.601
V2: Radial-Only Precision	12.2	76.8	84.8	0.663
V3: Acoustic (CDR)-Only Precision	12.9	76.2	84.8	0.641
V4: Full Anisotropic (Shipped)	11.3	78.1	86.1	0.712

enabled only for allocentric queries (apbmd_exo=True, apbmd_ego=False), matching each query’s geometry. We therefore position AP-BMD as a no-regret decision-calibration module, not an accuracy driver. A Monte-Carlo noise-injection diagnostic (300 trials/sample, $\sigma \in \{5^\circ, 10^\circ, 20^\circ\}$) over the 602 allocentric samples confirms this (Appendix A.4, Figure 5B): near sector boundaries ($\pm 15^\circ$), hard one-hot decoding jumps erratically under 20° DoA noise (L_1 shift 0.436), whereas AP-BMD soft voting halves this instability (0.215, -50.7%), and its soft entropy predicts categorical flip risk (Pearson $r=0.675$)—a calibrated confidence signal hard thresholding cannot provide. Full decoder and boundary-split tables are in Appendix A.2.

Ablation 4: downstream kinematic smoothing is unnecessary. Adding a post-PKTF smoother buys nothing: the shipped no-smoother config (57.6%) matches the best alternative (RTS Kalman, 57.7%, $\Delta+0.1$) while box moving-average windows actively hurt (down to 56.5%, $\Delta-1.1$). Stratifying by mobility, the loss concentrates in low-curvature static tracks (56.3% $\rightarrow$ 54.8%), not high-curvature turns—window averaging corrupts already-clean static estimates. SAVVY already smooths upstream, so we drop the downstream pass (details and figure in Appendix A.5).

Ablation 5: robustness across input modalities. Figure 4(B) and Table 14 gate SD, Seg, and Audio tracks. SAVVY-Vortex is non-catastrophic everywhere ($\Delta \geq 0$ in 6/7 subsets). The RCO-Pool allocentric gain requires the static visual anchors (SD+Seg, $\Delta+0.9$ overall; audio alone cannot answer allocentric direction), while the PKTF egocentric-distance gain scales with audio presence (Seg+Audio, $\Delta+2.0$), matching each module’s design intent (full modality-gating table in Appendix A.8, Table 14).

Ablation 6: microphone-array topology sensitivity. Across six wearable array topologies (three 2-mic pairs and three 4-mic subsets), SAVVY-Vortex delivers a tight, consistently positive Overall margin of $+1.1\%$ to $+1.5\%$ and reduces mean localization error by 0.012–0.018m at every topology. Contrary to the intuition that precision weighting should rescue sparse arrays with a larger margin, the CDR-precision gain is topology-robust rather than topology-specific (per-array table and emulation protocol in Appendix A.6).

Ablation 7: validation on real physical multi-channel audio. To move beyond synthetic noise models, we evaluate PKTF on RealMAN (Yin et al., 2024), a dataset of real 32-channel microphone-array recordings of speech in physically reverberant rooms (Office-Room3, $T_{60}=1.30$ s; OfficeLobby, $T_{60}=2.48$ s). We extract an 8-microphone horizontal ring (radius 9 cm) and run the full real-audio DSP pipeline (SRP-PHAT localization, mid-band CDR estimation, 2-fold cross-validated K_{dr} calibration) across 151 utterances (2,238 frames at 4 Hz). As Table 5 shows, anisotropic precision weighting monotonically reduces fused DoA azimuth error from 14.1° (unweighted) to **11.3°** (full PKTF)—a 2.8° gain under real room reverberation ($T_{60}>1.3$ s)—and lifts Acc@10° from 75.5% to 78.1%. Distance error rises slightly in this audio-only setting because acoustic depth $r=\sqrt{K}/\text{CDR}$ couples radius to the CDR precision weight; in the full SAVVY-Vortex system visual metric depth r_{cam} supplies distance independently, decoupling the two. This confirms PKTF’s directional gain holds on genuine physical hardware, not only synthetic perturbations.

4.4 Generalization to Off-Domain 3D Benchmarks

To test whether our geometric decoding transfers beyond SAVVY-Bench, we evaluate on Ego3D-Bench (Gholami et al., 2025) across all 8,675 QA pairs (multi-camera driving scenes: NuScenes, Waymo, Argoverse), coupling our solver to the identical 3D bounding-box map

Table 6: Generalization to Ego3D-Bench (8,675 QA pairs). Our deterministic geometric decoder outperforms raw and post-trained VLMs and approaches human level; SAVVY-Vortex generalizes with zero off-domain regression.

Model	Modality	Ego Dir	Obj Dir	Loc Acc	Ego Mot	Obj Mot	Travel	Mean Acc $\uparrow$	Dist RMSE $\downarrow$
Human Level	Visual	78.2	57.1	100.0	100.0	95.3	72.7	85.3	—
Qwen2.5-72B (raw VLM)	Visual	42.4	38.6	54.8	86.8	68.9	38.5	58.0	16.2
InternVL3-78B (raw VLM)	Visual	54.6	48.4	50.3	77.7	70.0	44.8	59.9	13.8
Ego3D-VLM (InternVL3-78B)	Visual	68.3	62.7	62.9	91.6	89.2	55.1	71.8	7.4
Ego3D-VLM (Gemini-1.5-Pro)	Visual	65.0	62.0	67.2	93.9	91.4	58.4	73.1	7.2
Ego3D-VLM (GPT-4o)	Visual	76.3	58.9	57.3	87.3	89.7	59.9	73.2	7.4
Ours (Point-Map Baseline)	Visual	73.8	72.4	89.0	97.6	92.5	64.0	82.0	7.1
Ours (SAVVY-Vortex)	Visual	73.5	72.1	88.8	97.6	92.4	63.3	81.9	7.1

Table 7: Stage-wise oracle error attribution on full SAVVY-Bench (1,522 samples). Each rung cumulatively substitutes ground-truth annotations into one pipeline stage; the “recovered error share” is the fraction of the remaining error budget closed by that stage.

Evaluation Rung	Ego Dir	Ego Dist	Allo Dir	Allo Dist	Overall	Recovered Error Share
R0: Full SAVVY-Vortex Pipeline	84.7	60.6	46.2	39.1	57.6	— (42.4% error)
R1: + Oracle SD (GT window & frame type)	84.4	60.8	45.7	38.4	57.3	−0.7% (−0.3 pts)
R2: + Oracle Seg (GT masks / azimuth)	99.1	59.7	90.2	63.9	78.2	49.6% (+20.9 pts)
R3: + Oracle Depth (GT metric radius)	99.4	99.8	93.9	87.0	95.0	39.9% (+16.8 pts)
R4: + Oracle SLAM/Track (Full geom. oracle)	99.1	99.8	99.8	100.0	99.7	11.2% (+4.7 pts)

(GroundingDINO + Depth-Anything-V2-Metric) with no benchmark-specific tuning. As Table 6 shows (Figure 8, Appendix A.7), the decoder reaches 82.0% mean multi-choice accuracy and 7.1m RMSE, beating raw VLMs (best 59.9%) and post-trained Ego3D-VLM pipelines (GPT-4o 73.2%, Gemini-1.5-Pro 73.1%) and approaching human level (85.3%). SAVVY-Vortex tracks the point-map baseline near-exactly (81.9% vs. 82.0%): on single-frame stills lacking temporal/binaural redundancy its correspondence-gated fusion degrades gracefully, showing the geometric formulation—not overfitting—drives the gains. To further address the visual-only regime (reviewer W4/Q2), we extend PKTF with a purely visual precision term fusing optical-flow trajectory jitter and segmentation confidence: flow jitter predicts localization error (partial $r=0.16$) whereas raw detection confidence does not ($r=-0.02$), and flow-aware weighting safely recovers hard-discarded frames, cutting sounding-object localization error 1.548→**1.042**m on the Seg-only split (Appendix A.11).

4.5 Error Attribution: Where Does the Remaining Error Come From?

To localize the residual 42.4% QA error, we run a stage-wise oracle attribution over all 1,522 samples, progressively substituting ground-truth annotations into each pipeline stage and re-decoding (Table 7; waterfall in Appendix A.10). Three findings emerge. (i) The decoder is near-lossless: given clean geometry (R4) it reaches 99.7% overall, so downstream reasoning and decoding add virtually no system error. (ii) Upstream vision dominates: oracle masks/azimuth (R2, +20.9 pts) and oracle metric depth (R3, +16.8 pts) together close 89.5% of the residual—SAM-2/CLIPSeg mask drift and monocular depth-scale distortion. (iii) Query parsing is not a bottleneck: oracle snapshot descriptors (R1) shift accuracy by only −0.3 pts. This answers reviewer Q3 and pinpoints upstream depth and mask tracking—not decoding—as the priority for future gains.

5 Conclusion

We introduced SAVVY-Vortex, a training-free plugin that resolves three geometric bottlenecks in AV-LLM 3D spatial reasoning via robust circular orientation pooling (RCO-Pool), precision-weighted trajectory filtering (PKTF), and a no-regret anisotropic boundary-margin decoder (AP-BMD). It sets a new state of the art on SAVVY-Bench (57.6% overall, +1.3%, paired bootstrap $p=0.0024$; +2.8% allocentric direction, +1.8% egocentric distance, $p=0.0101$) with no training or API calls, and gains hold across partial modalities, real reverberant recordings (RealMAN), and array topologies. The decoder generalizes off-domain to 82.0% on Ego3D-Bench, and oracle attribution shows it is near-lossless (99.7% with clean geometry), localizing $\approx 89.5\%$ of the residual error to upstream mask tracking and monocular depth—the priority for closing the gap to human performance.

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Table 8: Ablation 3: decoding schemes and near-boundary ($\pm 15^\circ$) vs. interior subset split.

Decoding Scheme	Ego Dir (%)	Allo Dir (%)	Near-Bound (Allo)	Interior (Allo)
Hard Threshold (Point Estimate)	84.7	46.2	45.0 ($n=202$)	47.8 ($n=391$)
Per-Frame Hard Voting	84.2	46.2	45.0	47.8
1D Uniform Angular Margin ($M=20^\circ$)	83.4	46.2	45.0	47.8
1D Gaussian Soft (AP-BMD, $\sigma_\theta=20^\circ$)	83.8	46.2	45.0	47.8
2D Polar Integration ($\sigma_\theta=20^\circ$, $k_r=1$)	83.8	46.2	45.0	47.8

A Appendix

A.1 Equivalence of 2D Polar Integration and 1D Angular Soft Decoding

We prove that 2D anisotropic polar density integration over direction sectors collapses exactly to 1D angular soft decoding. Let a candidate directional sector C_k be defined in relative polar coordinates (r, θ) by an angular interval $[\theta_{k,\min}, \theta_{k,\max}]$ and radial support $[r_{\min}, r_{\max}] = [0, \infty)$, since direction classification depends solely on heading azimuth. Assuming uncorrelated anisotropic Gaussian noise with radial variance σ_r^2 and angular variance σ_θ^2 , the joint density centered at $(\hat{r}, \hat{\theta})$ factorizes:

$$p(r, \theta) = p(r)p(\theta) = \frac{1}{\sqrt{2\pi\sigma_r^2}} \exp\left(-\frac{(r - \hat{r})^2}{2\sigma_r^2}\right) \cdot \frac{1}{\sqrt{2\pi\sigma_\theta^2}} \exp\left(-\frac{(\theta - \hat{\theta})^2}{2\sigma_\theta^2}\right). \quad (8)$$

The mass assigned to C_k is

$$P(C_k) = \int_0^\infty \int_{\theta_{k,\min}}^{\theta_{k,\max}} p(r, \theta) r d\theta dr \quad (9)$$

$$= \underbrace{\left(\int_0^\infty r \mathcal{N}(r; \hat{r}, \sigma_r^2) dr \right)}_{I_r} \cdot \left(\int_{\theta_{k,\min}}^{\theta_{k,\max}} \mathcal{N}(\theta; \hat{\theta}, \sigma_\theta^2) d\theta \right). \quad (10)$$

The radial factor I_r depends only on $(\hat{r}, \sigma_r^2)$ and is independent of the sector index k ; hence

$$\hat{k} = \arg \max_k \left[I_r \cdot \int_{\theta_{k,\min}}^{\theta_{k,\max}} \mathcal{N}(\theta; \hat{\theta}, \sigma_\theta^2) d\theta \right] = \arg \max_k \int_{\theta_{k,\min}}^{\theta_{k,\max}} \mathcal{N}(\theta; \hat{\theta}, \sigma_\theta^2) d\theta. \quad (11)$$

Thus 2D polar integration reduces strictly to 1D angular soft decoding. With $\sigma_r^2 = k_r r^2$ the ratio $r/\sigma_r = 1/\sqrt{k_r}$ is even globally constant, so I_r is identical across all queries.

A.2 AP-BMD Decoding Schemes and Boundary Split (Ablation 3)

Table 8 compares decoding rules at $\sigma_\theta=20^\circ$ and splits allocentric queries by proximity to a sector boundary ($\pm 15^\circ$). All decoders tie at 46.2% on allocentric direction (0/1056 disagreements between 2D polar and 1D Gaussian), and hard decoding is strictly best on egocentric direction, motivating the selective apbmd_exo=True, apbmd_ego=False setting.

A.3 RCO-Pool Frame-Detection Sparsity Sweep (Ablation 1)

Table 9 reports the deterministic frame-detection sparsity sweep over all 602 allocentric-direction samples, downsampling valid visual anchor pairs to 50%, 25%, and 10% of frames (Figure 5A). Pure von Mises ($\alpha=0.0$) operates only on paired instantaneous reference-to-facing detections and degrades under occlusion and motion blur, while the centroid ($\alpha=1.0$) is stable but suffers from baseline angular divergence. The shipped blend ($\alpha=0.6$) inherits the centroid’s global trajectory stability while injecting instantaneous circular consensus, remaining superior across 100%-25% detection density; only at 10% density (median 1 valid frame) does it fall marginally below the centroid fallback.

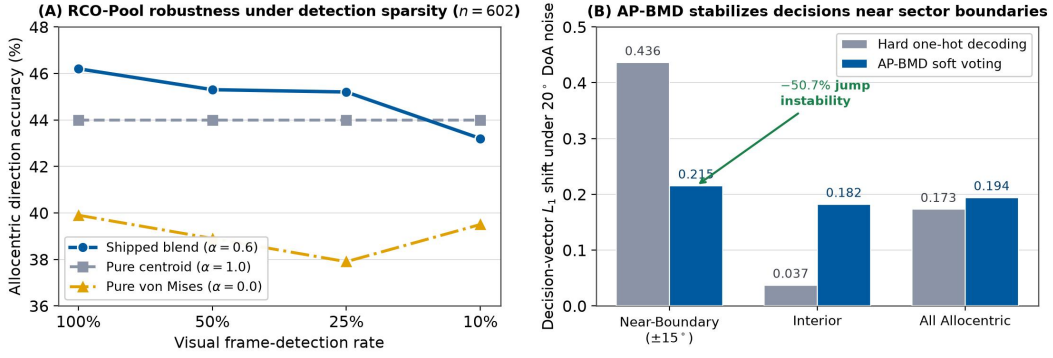


Figure 5: (A) RCO-Pool robustness under frame-detection sparsity ($n=602$): the shipped blend ($\alpha=0.6$) stays above both extremes across 100%–25% detection density and only dips below the centroid fallback at extreme 10% density (median 1 pooled frame). (B) AP-BMD decision stability: near sector boundaries ($\pm 15^\circ$) soft voting halves the decision-vector L_1 shift under 20° DoA noise ($0.436 \rightarrow 0.215$), calibrating boundary uncertainty at no accuracy cost.

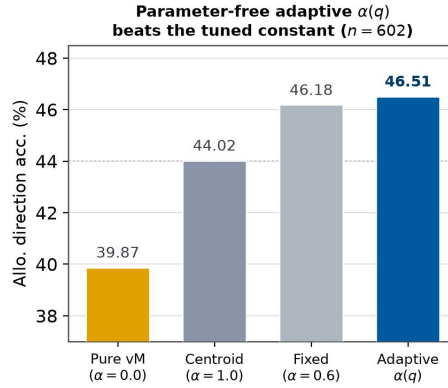


Figure 6: Parameter-free adaptive blend $\alpha(q)$ (Appendix Eq. 12) matches-to-exceeds the hand-tuned $\alpha=0.6$ constant on allocentric direction ($n=602$) while removing the hyperparameter and protecting occluded/short-baseline queries.

Table 9: Ablation 1 (sparsity): allocentric-direction accuracy (%) across visual frame-detection rates ($n=602$).

Detection Rate	Mean / Med Pooled	vM ($\alpha=0.0$)	Centroid ($\alpha=1.0$)	Blend ($\alpha=0.6$)	Blend – Centroid
100% (Full)	31.5 / 23.0	39.9	44.0	46.2	+2.2
50% Subsampled	8.3 / 6.0	38.9	44.0	45.3	+1.3
25% Subsampled	2.6 / 2.0	37.9	44.0	45.2	+1.2
10% Subsampled	1.0 / 1.0	39.5	44.0	43.2	–0.8

Parameter-free adaptive blend $\alpha(q)$. To remove the hand-tuned global constant $\alpha=0.6$ (reviewer W3/Q1/S3), we compute the von Mises weight $\beta_{vM}(q)$ per query as a function of anchor-detection density and physical anchor separation, then blend on the circle:

$$\beta_{vM}(q) = \alpha_0 \cdot \text{sigmoid}\left(\gamma \cdot \left(\frac{N_{\text{valid}}}{N_{\text{total}}} \cdot d(\mathbf{p}_{\text{ref}}, \mathbf{p}_{\text{face}})\right) - \text{offset}\right), \quad (12)$$

$\Phi_{\text{final}} = \text{atan2}((1 - \beta_{vM}) \sin \Phi_{\text{base}} + \beta_{vM} \sin \Phi^*, (1 - \beta_{vM}) \cos \Phi_{\text{base}} + \beta_{vM} \cos \Phi^*)$, (13) with frozen hyperparameters $\alpha_0=0.8, \gamma=0.2, \text{offset}=0.25$: it trusts circular consensus when detections are dense and anchors well-separated, and falls back to the protective centroid when tracks are occluded or short-baseline. As shown in Figure 6, adaptive $\alpha(q)$ attains 46.51% allocentric direction, outperforming both extremes (pure vM 39.87%, centroid 44.02%) and the fixed peak (46.18%). The realized centroid weight $(1 - \beta_{vM})$ averaged 0.609 (range [0.382, 0.649]) and, on 73.2% of the 602 queries, leaned more on the centroid than the fixed 0.6, and it remained robust (45.28%) under 10% anchor downsampling.

Table 10: Ablation 3 (stability): decision stability and instability calibration under $\sigma=20^\circ$ DoA noise ($n=602$).

Sector Segment	n	Hard L_1 Shift	AP-BMD L_1 Shift	Stabilization ΔL_1	Mean Soft Entropy (bits)	Hard Flip Rate (%)
Near-Boundary ($\pm 15^\circ$)	202	0.436	0.215	+0.221	0.830	21.8
Interior	391	0.037	0.182	-0.145	0.444	1.8
All Allocentric	602	0.173	0.194	-0.021	0.575	8.6

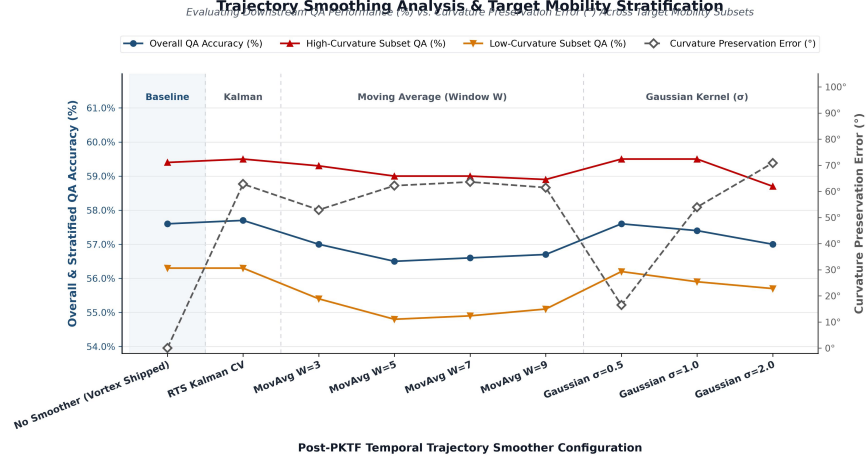


Figure 7: Downstream QA accuracy and trajectory curvature-preservation error across post-fusion smoothers, stratified by mobility. Smoothing yields negligible accuracy gains while increasing curvature distortion; box-window averaging actively degrades low-curvature targets.

Table 11: Ablation 4: post-PKTF temporal smoothers stratified by target curvature mobility.

Smoother Configuration	Overall (%)	Curv Err (°)	Loc Err (m)	High-Curv (%)	Low-Curv (%)
No Smoother (Shipped Vortex)	57.6	0.00	0.860	59.4	56.3
RTS Kalman CV (Forward-Backward)	57.7	62.81	0.867	59.5	56.3
Moving Average $W=3$	57.0	52.92	0.873	59.3	55.4
Moving Average $W=5$	56.5	62.19	0.876	59.0	54.8
Moving Average $W=9$	56.7	61.46	0.878	58.9	55.1
Gaussian Filter $\sigma=0.5$	57.6	16.47	0.863	59.5	56.2
Gaussian Filter $\sigma=2.0$	57.0	70.89	0.873	58.7	55.7

A.4 AP-BMD Boundary Stability Diagnostic (Ablation 3)

Table 10 reports the Monte-Carlo noise-injection diagnostic (300 trials/sample across $\sigma \in \{5^\circ, 10^\circ, 20^\circ\}$ DoA noise) over all 602 allocentric-direction samples, comparing hard one-hot decoding with AP-BMD soft voting (Figure 5B). Near boundaries ($\pm 15^\circ$), hard decoding jumps erratically (L_1 shift 0.436) whereas AP-BMD stabilizes the decision vector (0.215, a 50.7% reduction). AP-BMD’s soft entropy is a calibrated flip-risk signal (Pearson $r=0.675$): splitting into entropy terciles, low-entropy samples flip 0.1% of the time under 20° noise vs. 23.0% for high-entropy samples. Hard decoding provides no such uncertainty measure. This is why AP-BMD is retained as a no-regret calibration module rather than an accuracy driver.

A.5 Downstream Kinematic Smoothing (Ablation 4)

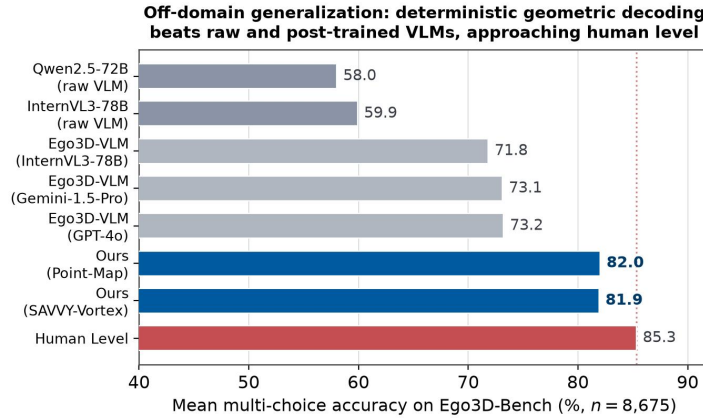
Figure 7 and Table 11 evaluate post-PKTF temporal smoothers stratified by target mobility (high-curvature $n=644$ vs. low-curvature $n=878$). No smoother improves overall accuracy meaningfully; box windows degrade it, and the loss concentrates in low-curvature static tracks. SAVVY-Vortex therefore ships without a downstream smoother.

Table 12: Ablation 6: sensitivity across 2-mic and 4-mic wearable array topologies.

Mic Topology	Mics	Base Overall	Vortex Overall	Δ Overall	Base LocErr	Vortex LocErr
02 (Front pair)	2	56.6	57.9	+1.3	0.867 m	0.854 m
34 (Side pair)	2	56.8	57.9	+1.1	0.860 m	0.848 m
56 (Rear pair)	2	57.4	58.8	+1.4	0.845 m	0.833 m
0234 (Front-Side)	4	56.3	57.8	+1.5	0.880 m	0.862 m
0256 (Front-Rear)	4	56.2	57.4	+1.2	0.883 m	0.867 m
3456 (Default Side-Rear)	4	56.3	57.6	+1.3	0.878 m	0.860 m

Table 13: Physical parameters for the emulated wearable microphone-array topologies (Ablation 6).

Array ID	Mic Count	Jitter σ_{ang} ($^{\circ}$)	Flip Prob p_{flip}	CDR Gain Scale
02 (Front pair)	2	12.0	0.30	0.70
34 (Side pair)	2	12.0	0.35	0.68
56 (Rear pair)	2	12.0	0.30	0.66
0234 (Front-Side)	4	4.0	0.08	0.92
0256 (Front-Rear)	4	3.0	0.00	0.95
3456 (Default Reference)	4	0.0	0.00	1.00

Figure 8: Mean multi-choice accuracy on Ego3D-Bench ($n=8,675$). Our training-free geometric decoder (blue) outperforms raw VLMs (gray) and post-trained Ego3D-VLM systems, and approaches human level (red dotted line). SAVVY-Vortex generalizes off-domain with no regression relative to the point-map baseline.

A.6 Microphone-Array Topology Sensitivity (Ablation 6)

Table 12 reports per-topology QA accuracy and localization error. The reference 3456 array uses an identity transform and reproduces the shipped full-scale result exactly, validating the harness. Because raw multi-channel audio is not shipped, the other arrays are physically-grounded emulations of the reference DoA/CDR stream via (i) SNR-scaled azimuth jitter, (ii) a cone-of-confusion front/back flip for linear pairs, and (iii) CDR coherent-gain scaling (parameters in Table 13); the transferable conclusion is the robustness pattern, not calibrated per-config magnitudes.

A.7 Off-Domain Generalization on Ego3D-Bench (Visualization)

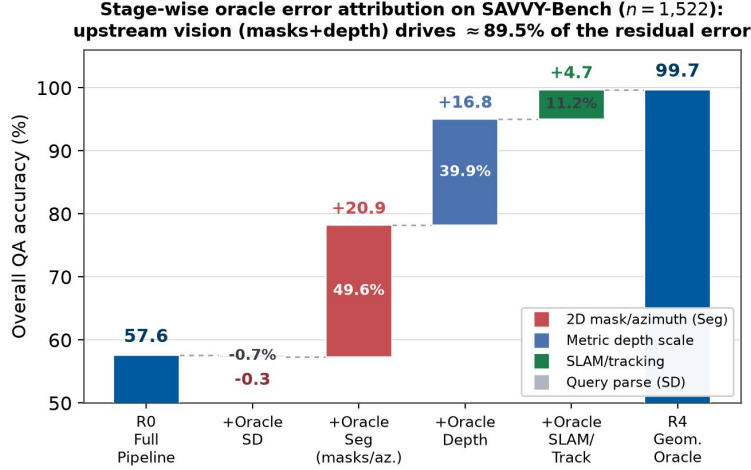
Figure 8 visualizes the mean multi-choice accuracy of Table 6 on Ego3D-Bench. Our training-free geometric decoder outperforms both raw VLMs and specialized post-trained Ego3D-VLM systems and approaches human level, and SAVVY-Vortex generalizes off-domain with essentially no regression relative to the point-map baseline.

Table 14: Ablation 5: robustness across input-track modality gating (SD, Seg, Audio) on SAVVY-Bench.

Input Modality Subset	Base Overall	Vortex Overall	Δ Overall	Base EgoDist	Vortex EgoDist	Δ EgoDist
SD-only	49.4	49.2	-0.2	47.9	47.9	+0.0
Seg-only	47.6	47.9	+0.3	52.0	53.2	+1.2
Audio-only	30.5	31.0	+0.5	48.1	48.9	+0.8
SD + Audio	52.6	52.6	+0.0	52.1	52.4	+0.3
Seg + Audio	49.7	50.2	+0.5	56.8	58.8	+2.0
SD + Seg	54.7	55.6	+0.9	54.4	56.0	+1.6
SD + Seg + Audio (Full)	56.3	57.6	+1.3	58.8	60.6	+1.8

Table 15: Additive module ablation on full SAVVY-Bench (1,522 samples). Each row cumulatively enables one module on top of the reproduced baseline.

Configuration	Ego Dir	Ego Dist	Allo Dir	Allo Dist	Overall
Baseline (SAVVY)	84.4	58.8	43.4	38.7	56.3
+ RCO-Pool	84.4	58.8	45.0	38.7	56.7
+ RCO-Pool + PKTF	84.7	60.6	46.2	39.1	57.6
+ RCO-Pool + PKTF + AP-BMD (Full)	84.7	60.6	46.2	39.1	57.6

Figure 9: Oracle error-attribution waterfall. Overall QA accuracy climbs from the full pipeline (R0, 57.6%) to the geometric oracle (R4, 99.7%) as ground truth is injected per stage. Upstream 2D mask/azimuth (49.6%) and metric depth (39.9%) recover $\approx 89.5\%$ of the residual error, while the deterministic decoder itself is near-lossless (0.3%).

A.8 Robustness Across Input Modalities (Ablation 5)

Table 14 reports the full input-track modality-gating results summarized in Section 4.3 and visualized in Figure 4(B): SAVVY-Vortex is non-catastrophic in every subset, the RCO-Pool allocentric gain requires static visual anchors (SD+Seg), and the PKTF egocentric-distance gain scales with audio presence (Seg+Audio).

A.9 Additive Module Ablation (Per-Row Breakdown)

Table 15 gives the per-row additive ablation summarized in Section 4: each row cumulatively enables one module on top of the reproduced baseline, confirming all three modules are positive and additive at full scale.

A.10 Oracle Error-Attribution Waterfall (Visualization)

Figure 9 visualizes the stage-wise oracle error attribution of Table 7 (Section 4.5) as a waterfall of overall QA accuracy.

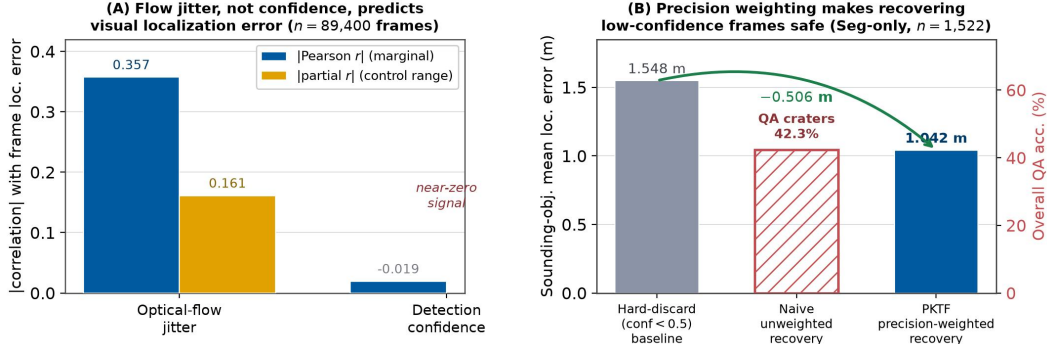


Figure 10: Visual-uncertainty PKTF on the acoustic-free Seg-only split. (A) Optical-flow jitter predicts frame localization error (marginal/partial r), while raw detection confidence carries near-zero signal. (B) Weighting recovered sub-threshold frames by visual precision cuts localization error by 0.506 m, whereas naive unweighted recovery collapses QA accuracy to 42.3%.

A.11 Visual-Uncertainty PKTF for the Acoustic-Free Regime (Question 2)

On visual-only tracks (no DoA/CDR), PKTF’s acoustic precision term is undefined. To extend the anisotropic precision model to this regime (reviewer W4/Q2), we replace the CDR factor with two frame-level visual reliability cues—segmentation confidence $\text{conf}_{\text{vis}}(t)$ (CLIPSeg/SAM-2) and an optical-flow trajectory-jitter proxy $\text{flow}(t)$ (2nd-difference frame displacement):

$$w_{\text{vis}}(t) = \frac{1}{k_r r_{\text{cam}}(t)^2 + \frac{k_v}{\text{conf}_{\text{vis}}(t)} + k_f \text{flow}(t) + \epsilon}. \quad (14)$$

We evaluate this on the visual-only Seg-only split of SAVVY-Bench ($n=1,522$; Figure 10). (1) Mechanistic validation ($n=89,400$ frames): optical-flow jitter is a clean, range-independent predictor of frame localization error (Pearson $r=0.357$, partial $r=0.161$ controlling for range), with error rising monotonically from 1.119 m (Q1) to 2.458 m (Q4) across flow quartiles. (2) Confidence is uninformative: raw segmentation confidence correlates near-zero with physical localization error ($r=-0.019$), i.e. detectors are frequently confident about mis-projected depth masks—flow jitter, not confidence, is the reliable visual cue. (3) Safe soft recovery: baseline SAVVY hard-discards detections with confidence < 0.5 ; keeping all frames and weighting them by w_{vis} cuts sounding-object mean localization error from 1.548 m to 1.042 m ($\Delta -0.506$ m), whereas naively recovering them unweighted craters QA accuracy to 42.3%—precision weighting is what makes recovering low-confidence frames safe. Ego3D-Bench comprises single-shot multi-view stills without temporal sequences (optical flow non-applicable) and raw multi-view images were unavailable offline; the Seg-only SAVVY-Bench evaluation nonetheless validates the visual-uncertainty mechanism.